

todo Co-Formation Games

Jens Friedel

*School of Engineering Mathematics
and Technology
University of Bristol
Bristol, United Kingdom
jens.friedel@bristol.ac.uk*

Richard Pettigrew

*Department of Philosophy
University of Bristol
Bristol, United Kingdom
richard.pettigrew@bristol.ac.uk*

Martin Bullinger

*School of Engineering Mathematics
and Technology
University of Bristol
Bristol, United Kingdom
martin.bullinger@bristol.ac.uk*

Abstract—This report proposes a novel model combining network creation studies and opinion dynamics. After introducing each Network Formation Games (NFGs) and Social Influence Models (SIMs), I compose a model allowing for co-evolution of opinions and networks and containing both NFGs and SIMs as special cases. Other sections consider basic dynamics, optimal states, and research directions of these Co-Formation Games (CFGs), as well as showcase an Julia module implementing CFGgames. Overall, the model offers a promising basis for future research projects, while broadening the perspective of NFGs and SIMs.

Index Terms—Collective dynamics, Game theory, Network formation games, Network theory (graphs)

I. INTRODUCTION

Network dynamics plays an ever-important role in describing the modern world, with connective plasticity and local optimizations enabling the emergence of global order. Since this framework can describe a vast variety of phenomena, researchers employed numerous approaches — often multi-disciplinary ones — to analyze these systems. Specifically, networks of human-like agents have been subject of studies ranging from mostly theoretical analysis to phenomenological social science works [1], [2].

Even as the sheer numerousness has inspired some authors to question each model’s marginal benefit, considering strangs of research showcases a lack of holistic perspectives [1]: As a subfield of game theory, Network Formation Games (NFGs) study how self-interested agents create and drive the structure of networks [3]. In contrast, Social Influence Models (SIMs) focus on inner states of nodes (opinions) diffusing through it. In the case of social networks, this raises a tension: As opinion propagation and network formation can both be dynamic and influence each other, results from studies regarding only NFG or SIM do not necessarily generalize to the case of simultaneous dynamics.

This is an important caveat, as real social network behaviour rarely reduces to the semi-static treatment implied by NFGs and SIMs. Yet, even though there are works trying to combine some aspects of the other’s influence, they often do so as a small part of the original model’s main mechanism (e.g., [4], [5], [6]). Moreover, there is still no work unifying both approaches in a way sheltering results from both disciplines.

This report explores a model which allows topology and opinions to influence each other, while including NFGs and

SIMs as special cases. After presenting NFGs and SIMs, I present and justify each building block of the model, before assessing some basic traits with regard to possible dynamics, optimality and a simulation module implemented in the Julia programming language.

Network creation studies stylistic models of how networks evolve, e.g., social networks. Opinion dynamics study how an opinion propagates through an existing network. The latter suggests that the network impacts the opinion dynamics. However, opinions may also influence what network connections form. The goal of the project to come up with and study a model where network creation and opinion propagation dynamically influence each other. The aim is to understand the co- evolution of opinions and networks. Dependent on the interest of the student, this question could be studied theoretically (e.g., describe the structure of stable networks that evolve) or computationally (through simulations).

II. METHODS

A. Definitions & Equations

TABLE I
LIST OF VARIABLES USED IN THIS DOCUMENT

Variable	Symbol	Domain
Number of nodes	n	$\mathbb{N}$
Node of concern	i	$\{1, \dots, n\}$
Set of i ’s neighbours	K_i	$\mathbb{N}^{ K_i }$
Network graph ¹	G	$\{0, 1\}^{n \times n}$
Opinion	x_i	$\mathbb{R}$
Action profile	a_i	$\{0, 1\}^n$
Costs per edge	α_c	$[0, \infty[$
Pairwise distance	d_{ij}	$\mathbb{N}$
Network valuation	v	$] - \infty, 0]$
Network expenses	f^c	$] - \infty, 0[$
Cost function	c	$] - \infty, 0[$
Communication rate	r_c	$[0, \infty[$
Formation rate	r_f	$\{0, 1\}$
Convergence threshold	ε	$[0, \alpha_c]$

where a norm $\|\cdot\|$ is the L0 norm, returning the count of non-zero elements.

¹The graph is represented as an adjacency matrix.

TABLE II
THE PLANETS OF THE SOLAR SYSTEM AND THEIR AVERAGE DISTANCE FROM
THE SUN

Planet	Distance (million km)
Mercury	57.9
Venus	108.2
Earth	149.6
Mars	227.9
Jupiter	778.6
Saturn	1,433.5
Uranus	2,872.5
Neptune	4,495.1

a) *Equations:*

- Cost function

$$c_i = v_i(a) + f^c(a) \quad (1)$$

- Valuation function

$$v_i(a) = - \sum_{j=1, \dots, n} d_{ij} \quad (2)$$

- Network expenses:

$$f_i^c(a) = -\alpha_c \|a_i\| - \frac{1}{\|k\|} \sum_{j \in K_i} |x_i - x_j| \quad (3)$$

b) *Algorithm:*

```

1 Initialise  $t = 1, c$ 
2 while ( $t < t_{\max} \wedge \varepsilon < \Delta c$ )
3   if  $\text{mod}(t, r_c) = 0$  then
4     select random node  $i$ 
5      $x_i$  arrow.l get opinion update  $x_i^u$ 
6   end if
7   if  $r_f = 1$  then
8     select random node  $i$ 
9      $a_i$  arrow.l get action update  $a_i^u$ 
10  end if
11   $t += 1$ 
12   $c(t)$  arrow.l get new costs  $c_t$ 
13   $\Delta c = c_{t-1} - c(t)$ 
14 end while
```

Algorithm 1. Dynamics of our synchronous move game

III. OLD STUFF FROM THE TEMPLATE



Fig. 1. A circle representing the Sun.

In Fig. 1 you can see a common representation of the Sun, which is a star that is located at the center of the solar system.

In Table II, you see the planets of the solar system and their average distance from the Sun. The distances were calculated with (1) that we presented in Section II.

REFERENCES

- [1] P. Sobkowicz, “Whither Now, Opinion Modelers?,” in *Frontiers of Physics*, 2020. [Online]. Available: <https://api.semanticscholar.org/CorpusID:226981914>
- [2] A. Flache *et al.*, “Models of Social Influence: Towards the Next Frontiers,” *J. Artif. Soc. Soc. Simul.*, vol. 20, 2017, [Online]. Available: <https://api.semanticscholar.org/CorpusID:3433103>
- [3] A. Fabrikant, A. Luthra, E. N. Maneva, C. H. Papadimitriou, and S. Shenker, “On a network creation game,” in *ACM SIGACT-SIGOPS Symposium on Principles of Distributed Computing*, 2003. [Online]. Available: <https://api.semanticscholar.org/CorpusID:5263612>
- [4] M. Bullinger, P. Lenzner, and A. Melnichenko, “Network creation with homophilic agents,” *Social Choice and Welfare*, vol. 64, pp. 915–959, 2022, [Online]. Available: <https://api.semanticscholar.org/CorpusID:248476098>
- [5] F. Baumann, P. Lorenz-Spreen, I. M. Sokolov, and M. Starnini, “Modeling echo chambers and polarization dynamics in social networks,” *Physical review letters*, p. 48301, 2019, [Online]. Available: <https://api.semanticscholar.org/CorpusID:195750744>
- [6] A. Nigam *et al.*, “ONE-M: Modeling the Co-evolution of Opinions and Network Connections,” in *ECML/PKDD*, 2018. [Online]. Available: <https://api.semanticscholar.org/CorpusID:51695921>